

The Grushin topological-dimension conjecture of arXiv:1406.4025

Literature-status note for `fa_banach.001`

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Status. `literature_already_answered` (full affirmative).

Original conjecture

Chen and Ouhabaz study the Grushin operator

$$L = -\Delta_x - |x|^2 \Delta_y \quad \text{on } \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}.$$

Their Theorems 1.1 and 1.2 use

$$D = \max\{d_1 + d_2, 2d_2\} \quad \text{and} \quad p_{d_1, d_2} = \min \left\{ \frac{2d_1}{d_1 + 2}, \frac{2(d_2 + 1)}{d_2 + 3} \right\}.$$

In the unnumbered Conjecture on page 3 of the arXiv PDF, immediately after those theorems, they write that the previous theorems should hold with $D = d_1 + d_2$ instead of $D = \max\{d_1 + d_2, 2d_2\}$ [1]. Thus the open part asks for both:

1. the p -specific spectral multiplier threshold $s > (d_1 + d_2)(1/p - 1/2)$ in the source range; and
2. uniform L^p boundedness of $(1 - tL)_+^\delta$ when $\delta > (d_1 + d_2)(1/p - 1/2) - 1/2$ in the same range.

The source observes that the multiplier assertion was already known for $p = 1$; the unresolved regime was the sharp p -specific improvement for $p > 1$ when $d_2 > d_1$.

Answer

Niedorf explicitly identifies the expectation of Chen and Ouhabaz and cites their paper before stating his main results [2]. On page 3 of the arXiv PDF, Theorem 1.1 proves that for $1 < p \leq p_{d_1, d_2}$, every bounded Borel function with local Sobolev regularity

$$s > (d_1 + d_2)(1/p - 1/2)$$

defines a bounded spectral multiplier $F(L)$ on L^p . Theorem 1.2 on the same page proves that for $1 \leq p \leq p_{d_1, d_2}$ and

$$\delta > (d_1 + d_2)(1/p - 1/2) - 1/2,$$

the Bochner–Riesz means $(1 - tL)_+^\delta$ are uniformly bounded on L^p . These are exactly the two source theorems with D replaced by the topological dimension $d_1 + d_2$. Together with the already-known $p = 1$ multiplier endpoint recorded by the source, this is a full affirmative answer to the displayed conjecture.

The answering authors knew the relation: the abstract says their result improves the Chen–Ouhabaz theorem, and the introduction expressly attributes the expected replacement to Chen and Ouhabaz before stating Theorems 1.1 and 1.2.

Scope and search evidence

This packet concerns only the stated Chen–Ouhabaz range $1 \leq p \leq p_{d_1, d_2}$ and the two conclusions in their unnumbered Conjecture. It does not assert a wider p -range or resolve the classical Euclidean Bochner–Riesz conjecture.

The four run indexes were searched for arXiv:1406.4025 and the terms “Grushin”, “spectral multiplier”, and “Bochner–Riesz”; they contained related Grushin packets but no record for this exact conjecture. Exact arXiv and title/keyword searches located arXiv:2110.10058. The identification was then checked in the two PDFs: the source Conjecture is on page 3, and the answering discussion and Theorems 1.1–1.2 are on page 3 of the supporting paper. No new mathematical result is claimed.

References

- [1] P. Chen and E. M. Ouhabaz, *Weighted restriction type estimates for Grushin operators and application to spectral multipliers and Bochner–Riesz summability*, Math. Z. **282** (2016), 663–678; arXiv:1406.4025, <https://arxiv.org/abs/1406.4025>.
- [2] L. Nierdorf, *A p -specific spectral multiplier theorem with sharp regularity bound for Grushin operators*, Math. Z. **301** (2022), 4153–4173; arXiv:2110.10058, <https://arxiv.org/abs/2110.10058>.